

Getting Started with CS2316

This document contains instructions for installing the software used in this course.

Note: The terminology included in this document and a basic understanding of the installation process is course content therefore it may be included on exams or quizzes.

Python, like most coding languages, is constantly evolving with new modules being added and existing modules being modified. This handout provides a basic understanding; however, true mastery requires practice and your own exploration of what python has to offer.

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Installation Verification (HW00)

HW00 is designed to make sure you have all of the software necessary for the class: Python, MySQL Server, and MySQL Workbench. After completing the software installations necessary for the course detailed in this file, follow the steps below to confirm that everything works properly.

1. Download the `TestImports.py` file and save it to your CS2316 directory
2. Open a text editor and create a file named `hello_world.py`
3. Save the file in the Homework directory
4. Within the `hello_world.py` file type `print("Hello World!!")`
5. Save the file, `ctr+s` or `cmd+s`
6. Open the command line
7. Navigate to the CS2316 directory
8. Type `dir` (Windows) or `ls` (Mac/Linux) to verify the directory for Homework and the `TestImports.py` file is present
9. Change to the Homework directory (`cd Homework`)
10. Execute `conda -V` in the command line to verify that conda is recognized as a command. You should have a version that is either around conda 4.8.x (or higher) or conda 22.11.x (or higher)
11. Execute `python -V` in the command line to verify that python 3.7.x or higher is installed and that python is recognized as a command
12. Execute `ipython -V` in the command line to verify that ipython is recognized as a command. You should have a version that is around ipython 7.3.x
13. Run your program by executing `python hello_world.py` from the command line
14. Run the `TestImports.py` file by executing `python TestImports.py` from the command line
15. Open up MySQL Workbench and connect to your local instance
16. Type and run the following command in MySQL Workbench: `select 'CS 2316!';`
 - NOTE: This command might error. All this is testing is whether or not the software is working. Please don't be concerned by the error.

Finally, for **the first homework of the semester (HW00)**, you must submit a video depicting the above steps before the due date listed on the living schedule to demonstrate that you know how to navigate the Command Prompt or Terminal and that your computer has the proper software installed. Please keep this video under 5 minutes and make sure to watch it to see that it actually shows up before submitting. **You do not need to do every step live, but you need to show proof of each being FULLY COMPLETED in order to receive credit.**

Introduction

Hey there! Welcome to CS2316, Data Input and Manipulation for Engineers. In order to begin manipulating data, we wanted to create a centralized space for the common information we will be using throughout the semester that you can easily refer back to. In this guide, you will find terminology, installation guides for our essential software, setup guides in every flavor of OS (Windows, MacOS, and Linux), and even your **first Homework assignment of the semester!**

CS2316 is a computer heavy course, and requires you to be very familiar with your particular computer and operating system. We have listed a couple of things that we feel are necessary for you to have a successful semester:

- 1) **MOST IMPORTANT:** Please make sure you have all of the softwares necessary for the semester installed by the end of the second week.
- 2) Please try to have your computer as up to date as possible at the start of the semester. We recommend that you do not update your computer throughout the course of the semester as this may cause certain softwares to stop working.
- 3) Finally we recommend that you keep your computer very organized with a neat file structure and naming system. This will make locating and running files much easier throughout the course of the semester. Additionally you are much less likely to lose a file if your computer is organized.

Remember computers are a little like cats. They all behave relatively similarly, but each has its own weird quirks. While the TAs are familiar with all operating systems, each computer is set up differently meaning that each problem is potentially unique. The TAs are available to help you with any questions you may have regarding installation, but it is up to you to know how to operate your computer.

Terminology

Conda - A distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment for Windows, Linux, and MacOS

File - A file is an object on a computer that stores data, information, settings, or commands used with a computer program

File Extension - The characters listed after a period in a file name (e.g. .py, .ipynb, .txt, .csv, etc.)

Directory - A folder like structure that contains files or directories

Command Line - A program used to access directories and execute commands (Terminal for Mac and Linux and Command Prompt for Windows)

Text Editor - A computer program that allows users to enter, change, and store contents within a file

Python Interpreter - A computer program that reads and runs snippets of code and can be accessed from the command line or python shell

Shell - An interactive environment used to access an interpreter, for this course either the python or MySQL interpreter

Path - A unique location in a file system (e.g. C:/user/CS2316/Homework)

Environment Variable - A variable whose value is defined by the environment (typically the operating system) in which it is running

Package Manager - A collection of software tools that automates the process of installing, upgrading, configuring, and removing computer programs for a computer's operating system

Source Code - A computer program as it is originally written (i.e., typed into a computer) by a human in a programming language

Open Source - Software with publicly available source code, i.e. non-proprietary software

Module - A python file that can be accessed in other programs

Pip - The standard package manager used to install and manage software packages written in Python (in this course we do not use pip, we use conda instead)

Relational Database - Data in a relational database is organized into tables of columns and rows.

Relational Database Management System - A program that allows you to create, update, and administer a relational database. Most relational database management systems use the SQL language to access the database. Popular relational database management systems include MySQL, PostgreSQL, Oracle DB, SQL Server, and SQLite. We will be using MySQL in this class.

File Structure

To create consistency we require that all students have an organized and relatively similar directory for this course. The barebones file structure should look like...

User

...

CS2316

Homework

Participation

In other words, the paths “user/.../CS2316/Homework” and “user/.../CS2316/Participation” should be accessible on your machine. The names do not have to match, the structure is what matters. It may be beneficial to add subdirectories like Recitation, Lecture, Notes, etc. to the CS2316 directory as needed.

Command Line Crash Course

You have probably noticed that your file system is structured like a tree. Just like how you can navigate through your file system by clicking around on an interface like File Explorer or Finder, you can do the same navigation through the command line. The term command line is used to reference the Command Prompt (Windows) or Terminal (Mac/Linux). To open up Command Prompt (Windows), search and run **cmd.exe** (windows key+r, cmd, enter). To open up Terminal (Mac/Linux), search and run **Terminal**

Some Basic Commands

`mkdir directory_name` - creates a directory titled `directory_name`

`cd sub_directory_name` - change directory.

`cd ..` - go back to the parent directory of the current directory.

`ls` - list the content in the current directory (Linux or Mac)

`dir` - list the content in the current directory (Windows)

`cat` - display the content of the file (Linux or Mac)

`type` - display the content of the file (Windows)

Make sure to play around and get comfortable with them!

More on cmd line:

<http://matt.might.net/articles/basic-unix/> (Linux or Mac)

<https://www.computerhope.com/issues/chusedos.htm> (Windows)

Python Environment Installation

In order to run a python program, the Python interpreter must be accessible on your machine, specifically through the Command Line or Terminal. Instead of installing Python from python.org, we **require** the installation of the package manager Conda to make our lives easier in the future.

Package Manager: Conda

Conda is an open-source package management system and environment management system that runs on Windows, macOS and Linux. In other words, it is a package manager that helps you find, install, and update the packages you need from the Command Line or Terminal. Where do we get those packages? From Anaconda! Therefore, we need to install Anaconda, the robust version of conda, or Miniconda, the condensed version of conda.

More on Conda here: <https://conda.io/en/latest/>

Before starting the installation, you should be familiar with the differences between Anaconda and Miniconda.

Anaconda

Anaconda is a package manager, an environment manager, a Python/R data science distribution, and a collection of over 1,500+ open source packages. Over 250 popular data science packages are automatically installed with Anaconda, including all the packages we will be using in CS2316. For this reason, Anaconda would occupy approximately 3.2 GB of storage. Additional open-source

packages can be further installed from the Anaconda repository with `conda install package_name`.

Anaconda documentation: <https://docs.anaconda.com/anaconda/>

If you are not worried about your system's storage capacity, **we recommend you install Anaconda**, as the pre-installed packages will be useful for further ISYE courses. However, if you are facing a system with a very limited storage capacity, it is recommended you acquire Miniconda and only install the required packages separately. If you decide to install Miniconda, you should investigate its documentation for installation purposes.

For this course, we require you to install a Python version equal or greater than 3.7 and the conda package manager. Both are included in Anaconda and Miniconda, so you do not have to install both. Follow the installation guides corresponding to your system's OS in the following sections.

macOS Installation

Anaconda

If you decide to install Anaconda, you can follow the instructions listed in this section or from Anaconda's official macOS installation documentation:

<https://docs.anaconda.com/anaconda/install/mac-os/>

1. Navigate to <https://www.anaconda.com/products/individual#macos>, click "Skip Registration", and select the "Download" button. You should then see two download links for MacOS: 64-Bit Graphical Installer and 64-Bit Command Line Installer.
 - a. The 64-Bit Graphical Installer is a GUI-based installation that will guide you throughout the process. We recommend you use the Graphical Installer over the Command Line Installer. Make sure you download the package listed in Python 3.7.
 - b. If you have an Apple Silicon processor, M1 and M2, please scroll to the bottom of <https://www.anaconda.com/products/distribution#macos> and install the version for [64-Bit \(Apple Silicon\) Graphical Installer \(704.7 MB\)](#)
2. After the file has finished downloading, double-click the file to launch the installer and click "continue" to start the installation.
3. Read through the Introduction, Read Me, and License sections and answer the prompts.
4. Once you reach the Installation Type section, the installer will tell you how much space the installation will take and whether you want to customize your Install Location. We recommend you to install in the default location and ignore changing the installation path. Press "install".
5. If you are directed to the Destination Select section, choose "Install for me only" and click "continue".
6. The installer will then install Anaconda, along with Python 3.8.x, conda, and the other pre-installed packages. If the installation was successful, you should be taken to the

PyCharm IDE section. If the installation fails, close the installer and retry the process. If the error persists, come to office hours or submit a post on Ed to resolve this issue.

7. At the PyCharm IDE section, the installer will introduce you to PyCharm. Since we will not be utilizing any IDEs for this course, do not install PyCharm and click “continue”.
8. Once at the Summary section, you should see a message saying you have successfully installed Anaconda. Close the installer and verify you have the correct conda and python versions.
9. Open your Terminal (command + space, type in Terminal) and run the following commands:
 - a. `conda -V` (should have either conda 4.8.x or higher or conda 22.11.1 or higher)
 - b. `python -V` (should have Python 3.7.x or higher)

Note: If you are receiving Python 2.7 or receiving the message “command not found: python” and have installed Anaconda successfully, try running `python3 -V`. If you receive the “command not found” error or still displays Python 2.7, try running `conda activate base` and rerun `python -V`. If you still receive the same result, contact a TA immediately to resolve this issue.

This problem has been known to occur for users on macOS 10.15.x (macOS Catalina).

Common Errors:

For Mac OS, M1 or M2, scroll down to the bottom of the Mac download options and select the Graphical Installer M1 selection.

10. After you have verified the correct versions of conda and Python have been installed, run the following commands in Terminal:
 - a. `conda update conda`
 - b. `conda install requests pyqt beautifulsoup4 iPython numpy pandas pymysql matplotlib jupyter selenium`

If you receive any errors, please contact a TA as soon as possible.

You have just successfully updated and installed the software required for this course.

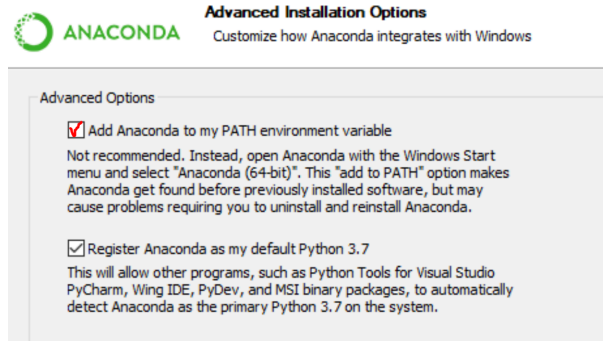
Windows Installation

Anaconda

If you decide to use Anaconda, follow the installation guide provided by Anaconda’s documentation:

<https://docs.anaconda.com/anaconda/install/windows/>

At step 8, check the box to add Anaconda to the PATH environment variable, as shown below.



After installation, open up the command line. Type in `conda -V` and `python -V`. You should get something like `Conda 4.8.x` or `Conda 22.11.1` or higher and `Python 3.8.x` or higher. If an error is thrown indicating that `conda` or `python` is not recognized visit the Environment Variables subsection for further details. Once `conda` is recognized as a command, execute the following to update and install the software used for this course:

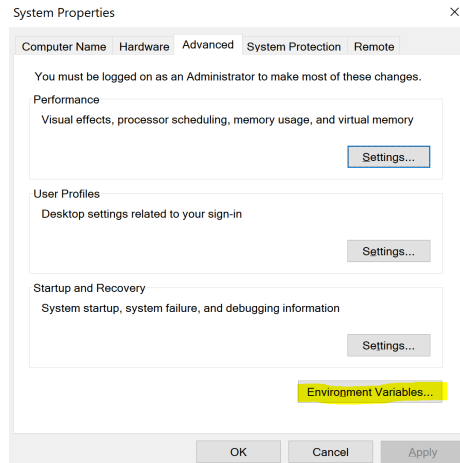
- `conda update conda`
- `conda install requests pyqt beautifulsoup4 iPython numpy pandas pymysql matplotlib jupyter`

Environment Variables

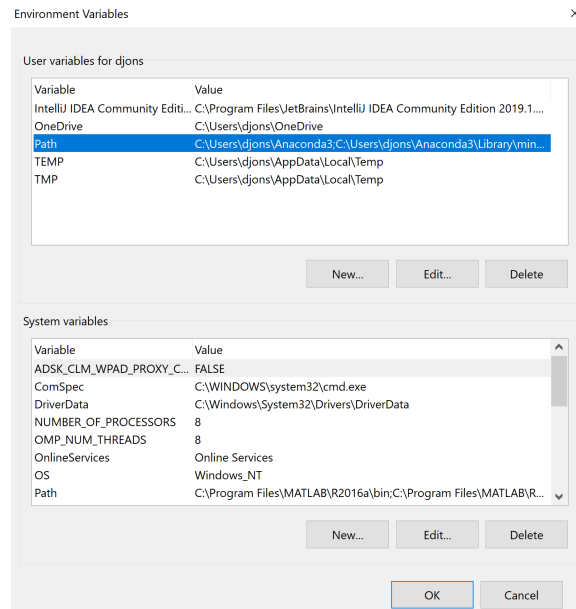
The environment variables tell your operating system which commands are recognized when executing commands in Command Prompt. When installing software like, `conda`, `python`, `java`, etc. it is vital that the path that holds the interpreters, compilers, or packages are saved as an environment variable. Why is this important? To avoid errors like this ...

```
C:\Users\djons\School\GT\Spring2020\CS2316\Participation>python hello_world.py
'python' is not recognized as an internal or external command,
operable program or batch file.
```

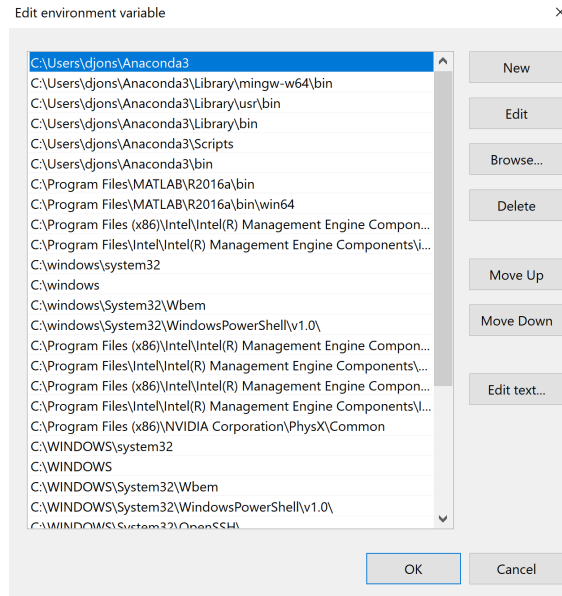
If you receive an error similar to the one above **you should add the path that holds the Python interpreter** to the environment variables. Before doing this, we should explore what paths are included in the environment variables. To view and edit the environment variables press the windows key and type “Environment Variable” then press enter. The following screen should appear:



Click the button highlighted in yellow, then a similar screen should appear:



Click on the variable Path (highlighted in blue) within the section “user variables”, then a similar window should appear:



This screen displays all the paths that my system recognizes as an environment variable. Note there are additional paths not applicable to this course. If you followed the installation instructions properly, similar paths should exist for Anaconda or Miniconda.

To get your system to recognize python as a command, the path to the interpreter and some other folders should be added. If a path containing Anaconda3 or Miniconda3 is not listed, then the path to the conda folder and the bin folder should be added. For example, adding “**C:\Users\<insert username>\Anaconda3**” and “**C:\Users\<insert username>\Anaconda3\bin**” may fix the problem. To see if this fixes the problem, save the results by selecting ok then open a new instance of the Command Prompt and type **python -V**, the resulting output should be a version of Python greater than 3.7. If you still receive the same result, come to office hours or post on Ed.

Note: The purpose of this section is to help with any software installation errors and is not testable material.

Linux Installation

Anaconda

If you decide to install Anaconda, you can follow the instructions listed in this section or from Anaconda's official macOS installation documentation:

<https://docs.anaconda.com/anaconda/install/linux/>

1. Navigate to <https://www.anaconda.com/products/individual#linux> and select the "Download" button. You should then see two download links for Linux: 64-Bit (x86) Installer and 64-Bit (Power8 and Power9) Installer. Download the file that is compatible with your system's OS.
2. Open up a terminal and run the following command:

```
bash ~/Downloads/Anaconda3-2020.02-Linux-x86_64.sh
```

 - a. If you did not download to your Downloads directory, replace `~/Downloads/` with the path to the file you downloaded.
3. The installer will then prompt you to review the license agreement terms. Press ENTER to view the license terms and enter "Yes" to agree.
4. The installer will prompt you to select your install location. We recommend you to use the default install location, so click "Enter".
5. The installer displays "PREFIX=/home/<user>/anaconda3" and continues the installation. It may take a few minutes to complete.
6. The installer will ask "Do you wish the installer to initialize Anaconda3 by running conda init?". Enter "yes".
7. After the installer finishes, it should display "Thank you for installing Anaconda3!". If you receive an error, try the installation process again. If you still receive the same result, come to office hours or submit a post on Ed to resolve this issue.
8. Skip over the section regarding PyCharm. We will not be using any IDEs for this course, so you do not need to install it.
9. Close and reopen your terminal window for the installation to take effect. You can enter the command `source ~/.bashrc` as an alternative.
10. To control whether or not each shell session has the base environment activated or not, run `conda config --set auto_activate_base false` or `true`. To run conda from anywhere without having the base environment activated by default, use `conda config --set auto_activate_base false`. This only works if you have run conda init first.
11. Open your Terminal and run the following commands:
 - a. `conda -V` (should have conda 4.8.x or higher)
 - b. `python -V` (should have Python 3.7.x or higher)

If you encounter any errors, come to office hours or submit a post on Ed to resolve this issue.

Installing a Text Editor

We will not be using an integrated development environment (IDE) such as IDLE, PyCharm, or IntelliJ in this class. Instead, we will be using the command line and text editors. **This does not mean that you cannot use an IDE, only that TAs are not liable to understand your preferred IDE completely.**

Text editors are basically just “notepads” where you can only edit plain texts. Plain texts are the opposite of rich texts, which consist of both character representations and the formatting of characters. Text editors come with more advanced features than notepads. For example, they do syntax highlighting automatically, which makes programs easier to read and write.

In this class, we will use Sublime Text. You can download it here: <http://www.sublimetext.com/>. You are welcome to use other popular text editors too, however we will not help troubleshoot problems as Sublime Text is the text editor we endorse.

Customizing Text Editors

To be a good programmer, you need to write code that is clear and consistently formatted. There are some configurations on text editors that will help you do that. We suggest that you configure your text editors to:

- Indent with spaces instead of TABs
- Set indent (tab) width to 4

The two specifications above can save you lots of frustrations! As a general rule of thumb, never mix spaces and TABs when you program in Python, even though they look the same. This is because Python uses indentation to separate sections of code and sometimes if TABs and spaces are mixed the interpreter gets confused and gives an ambiguous error. For further precaution and good practice, we recommend the following settings and visuals,

- Trim trailing white space at the end of lines
- Add a newline character to the end of files
- Show a visual ruler at 80 columns
- Show line numbers

The following section will show you how to enable these settings in sublime text.

Customizing Sublime Text

macOS: In Sublime Text’s application menu, select the Preferences menu, then click Settings - User.

Linux and Windows: In the main menu, select the Preferences menu, then Settings.

Two editor tabs will open side by side. The left contains the default settings, the right tab contains your customizations. The first time you open your settings the custom settings tab (on the right) will contain a blank JSON dictionary (curly braces). Replace the empty dictionary with the following:

```
{
```

```

    "tab_size": 4,
    "translate_tabs_to_spaces": true,
    "rulers": [80],
    "trim_trailing_white_space_on_save": true,
    "ensure_newline_at_eof_on_save": true
}

```

Finally, save the file and restart sublime text.

How to Write and Run a Python Program

1. Open sublime text and write your program.
2. Save the file with a .py extension in the desired directory.
3. Open up your command line. Navigate to the directory that your Python file is in (See commands above).
4. Type in `python file_name.py`
5. Read the error message if there's any and debug.

iPython

iPython is an enhanced shell for Python that allows for interactive computing. It is an environment that permits direct interaction with the python interpreter. Therefore, you can use iPython to test and develop code before finalizing a script. It is a convenient tool for learning or just playing around with new concepts.

To enter the iPython shell, type `ipython` in the command line and press enter.

iPython allows you to input lines of code and process the results.

iPython also comes with a bunch of useful magic functions. For example, `%timeit` allows users to see how long it takes Python to execute a certain line or block of code.

```

>>> %timeit a = 1
>>> 13.4 ns ± 0.394 ns per loop (mean ± std. dev. of 7 runs, 100000000
loops each)

```

More on iPython and magic functions here:

<https://iPython.readthedocs.io/en/stable/interactive/tutorial.html>

MySQL Installation

Next, we will install the Database Management System (DBMS) MySQL, which will be used later in the course.

macOS Installation

We will be using Oracle's MySQL Community downloads in conjunction with MySQL Workbench. Navigate to <https://dev.mysql.com/downloads/> and download the following DMG Archive files: **MySQL Community Server** and **MySQL Workbench**. Please ensure that you are downloading the correct file for your device.

If your computer has an Intel processor please make sure you are installing the program with (**x86, 64-bit**). If your computer has an Apple Silicon processor in it, you should install the program with (ARM, 64-bit). In order to discern this, click the Apple button on the top left of your screen, and refer to the "About My Mac":



If you have an Apple M1 or M2 chip, you have an Apple Silicon processor. Install the following MySQL Workbench: [macOS \(ARM, 64-bit\), DMG Archive](#)

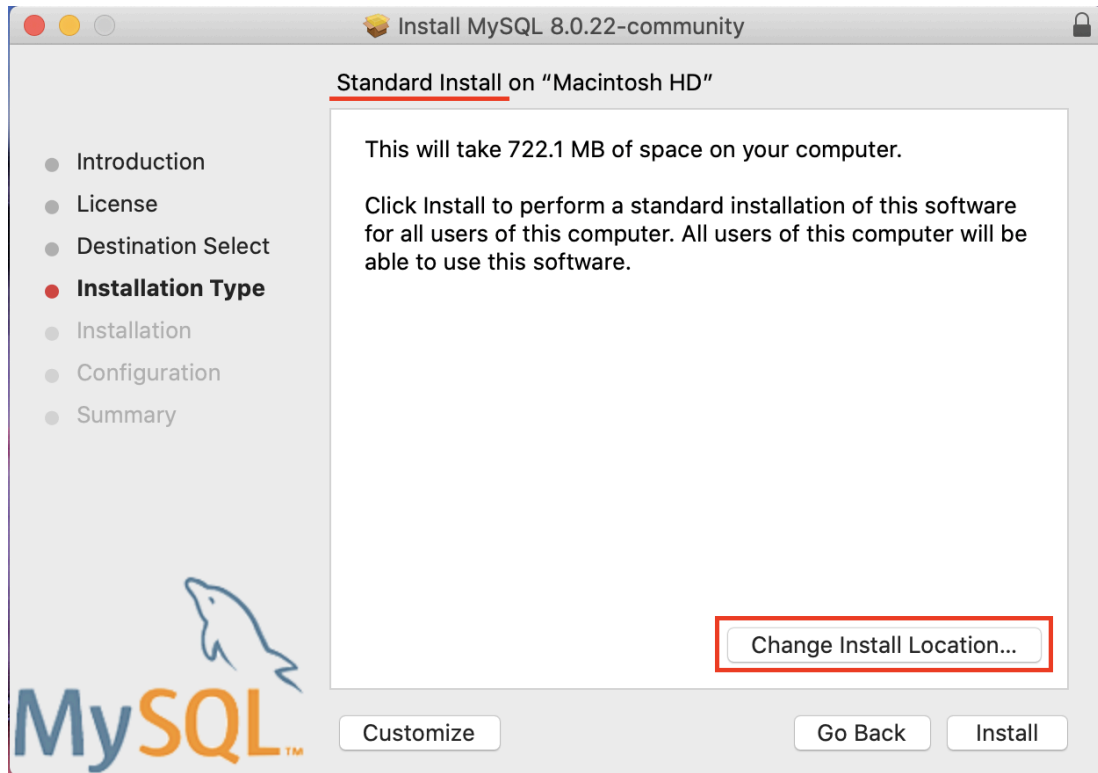
If you have an Intel processor, install the following MySQL Workbench: [macOS \(x86, 64-bit\), DMG Archive](#)

NOTE: If your computer is not up to date, we recommend updating your computer to the newest operating system before completing the install. If for whatever reason this is not possible, you will need to install an archived version of the workbench and server which can be found here: <https://downloads.mysql.com/archives/workbench/>

MySQL Community Server

Run the graphical installer once the **MySQL Community Server** DMG file is downloaded.

1. At the *Installation Type* page, make sure you are utilizing the Standard Install process.
2. The default Install Location should be set to "Install for all users of this computer". You can check the settings under the *Destination Select* page by clicking on the Change Install Location button.



3. Once you have confirmed the correct settings, continue the installation by clicking on the Install button.
4. After the installation has been completed, you will be greeted by the *Configuration* page, which will ask you to configure a password for your MySQL Server. For this class, we recommend you to use the **Strong Password Encryption**. Press Next.



5. You will then be prompted to enter a password for the “root” user. Please make sure you choose a password that you can easily remember, as you **will have to uninstall and reinstall MySQL** if you forget your root password. Uninstalling results in the loss of created databases, so please do not forget your password.

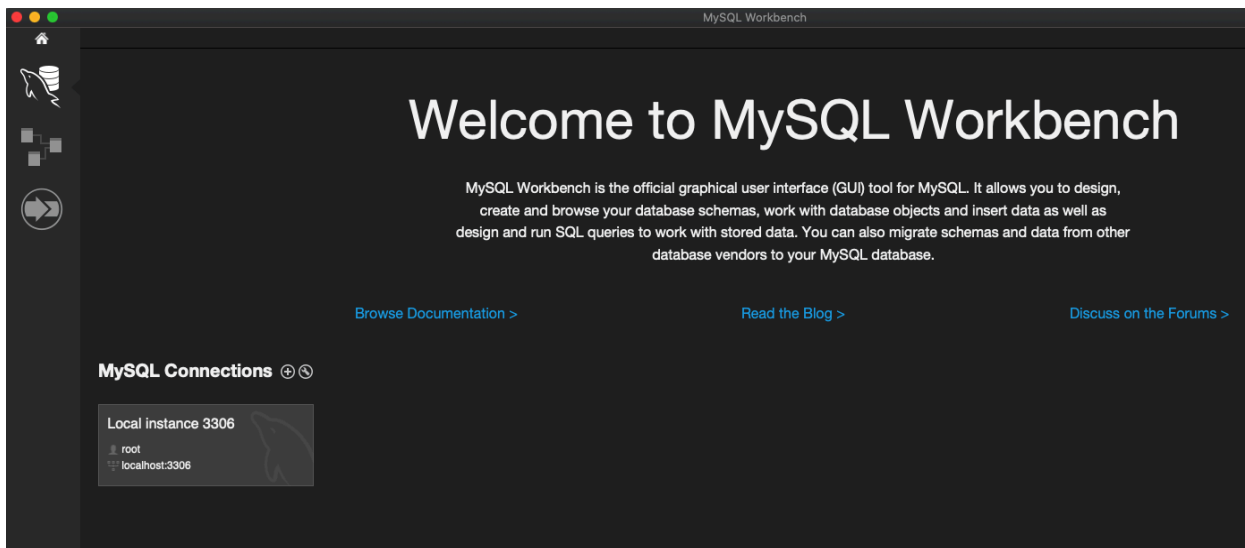
BE VERY CAREFUL! VERY IMPORTANT! Please go to the previous step and make sure that you **REMEMBER YOUR PASSWORD**. Write in your agenda or somewhere where you know you will be able to find it later :) If not, you will have to delete and redownload everything later in the semester.

6. Finish the installation once you have chosen your password. You should see that the MySQL Server has been successfully installed at the end screen.

MySQL Workbench

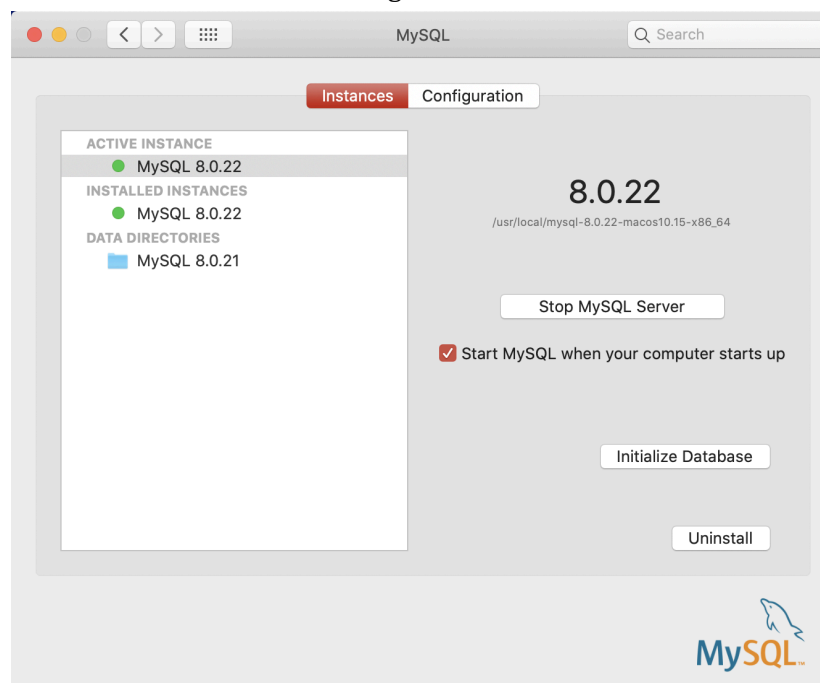
Run the graphical installer once the **MySQL Workbench** DMG file is downloaded.

1. Drag the MySQL Workbench application to the Applications folder.
2. Once the application has finished copying into the Applications folder, open your Applications folder by going to Finder > Applications.
3. Find “MySQLWorkbench” and right-click on the application. While holding the command key (⌘), press the “Open” button. Open up MySQL Workbench.
4. Once the MySQL Workbench window opens, you should see “Local instance XXXX” with the “root” user listed under **MySQL Connections**.



Accessing Local Instance

In order to access your local instance when connecting to MySQL, we must make sure that the MySQL server and instance are currently running. To check on the status of your instance, go to System Preferences > MySQL. If the color next to the instance is red, it is not currently running. Press on Start MySQL Server to run the instance. Once the color is green, exit out of System Preferences and restart MySQL Workbench. You can then connect to your Local instance by double-clicking on the “Local instance XXXX” region.



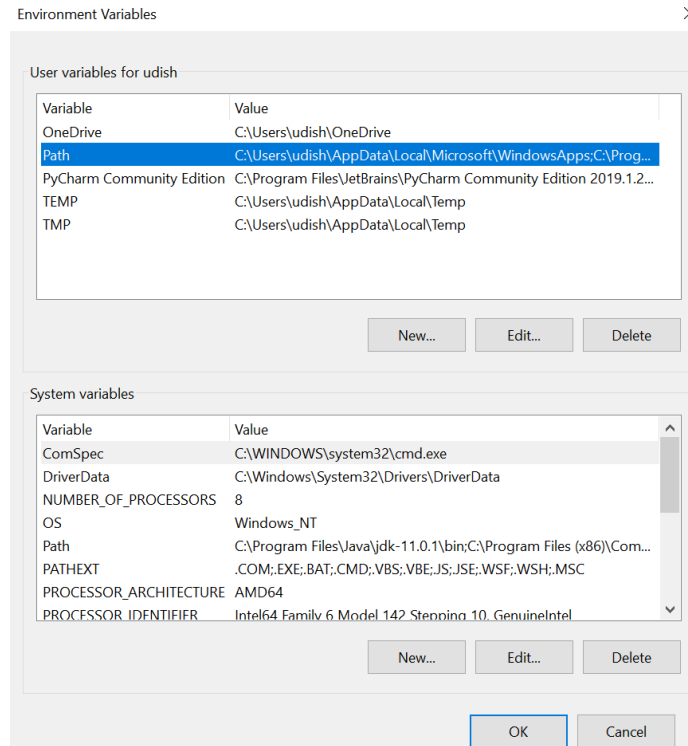
Windows Installation

Go to <https://dev.mysql.com/downloads/installer/>, select “MySQL Installer for Windows” at the bottom left, select the second installer (~300M), and download the installer. If prompted to make an account, you can select “No thanks, just start my download” at the bottom. **Make sure to follow the steps below *precisely* or else your MySQL may not install correctly.**

1. Select “Full” and click “Next”. If there are unsatisfied dependencies, click “Execute” before “Next”.
2. You should now see a list of products to be installed. Click “Execute”. If prompted, agree to terms and conditions.
3. Now you will be doing the Product Configuration.
 - a. Group Replication: Standalone MySQL Server/Classic MySQL Replication (This may not need to be configured. If there’s no prompt for this configuration, simply move on to Type and Networking)
 - b. Type and Networking: Make sure “Open Windows Firewall Port” is unchecked, leave all else as default.
 - c. Authentication Method: Use Legacy Authentication Method
 - d. Root Password: You may want to choose a **short and easy to remember password** (such as a basic PIN). After installation, you have the option of getting rid of it altogether. **If you forget this password, you may have to uninstall/reinstall MySQL later on, which will be very troublesome.**

BE VERY CAREFUL! VERY IMPORTANT! Please go to the previous step and make sure that you **REMEMBER YOUR PASSWORD**. Write in your agenda or somewhere where you know you will be able to find it later :) If not, you will have to delete and redownload everything later in the semester.

- e. Windows Service: Check Configure MySQL Server as a Windows Service, Check Start MySQL Server at System Startup, Select Run Windows Service as Standard System Account
 - f. Document Store: Uncheck it all, not needed (This may not need to be configured. If there’s no prompt for this configuration, simply move on to Apply Configuration)
 - g. Apply Configuration: Click execute (if fails, execute again), MySQL Router Configuration (Uncheck), Samples and Examples (enter root password you just made and click “Check” → Click “Next” → Apply Configuration, Click “Execute”
4. Finish Installation, Uncheck “Start MySQL shell”
5. MySQL workbench should open after finishing installation. If the window opens successfully you may minimize it for now.
6. Now, we will ensure that MySQL Server is in your PATH environment variables. **You must complete this step before visiting Office Hours for MySQL installation help.** Press the windows key next to your Fn and Alt key and type in “edit the system environment variables” → on the bottom right above the cancel button press “Environment variables”



A screen that looks like the one above should show up and you should click on the “Path” variable beneath the User Variables section and press “Edit”.

You should have a path that looks something like **C:\Program Files\MySQL\MySQL Server 8.0\bin**. If you have **C:\Program Files\MySQL\MySQL Shell 8.0\bin**, this is okay, you can simply replace “Shell” with “Server” and you should be good. Finally, after you verified that the **exact** path and spacing **C:\Program Files\MySQL\MySQL Server 8.0\bin** is added to the user environment variables click “OK” as many times as you are prompted.

To test that the mysql command is recognized by the Command Prompt, open a new Command Prompt window and execute **“mysql -u root -p”**. If you are prompted for your password, then congratulations you’ve installed MySQL correctly! Otherwise, if you get an error that says “mysql is not recognized as an internal or external command”, verify again that the **exact** path **C:\Program Files\MySQL\MySQL Server 8.0\bin** is listed under the user environment variables. If the environment variables are correct and mysql is still not recognized as a command please visit the TAs during office hours.

Linux Installation

For installing MySQL on Linux, follow the steps listed here:

<https://ubuntu.com/server/docs/databases-mysql>

1. Run the following commands in terminal:

```
sudo apt install mysql-server  
sudo service mysql status
```

2. After running the previous commands, install MySQL Workbench from the official website:

<https://dev.mysql.com/downloads/workbench/>

3. You should be able to install by double-clicking the file. If not, try installing it manually by running the following command in terminal:

```
sudo dpkg -i [Downloaded .deb file]
```

4. MySQL Workbench will not work by default. Run the following command in terminal:

```
mysql -u root -p
```

(if sudo is needed)

```
sudo mysql -u root -p
```

5. Once you are connected to mysql in terminal, follow the steps listed here:

<https://askubuntu.com/a/793750>

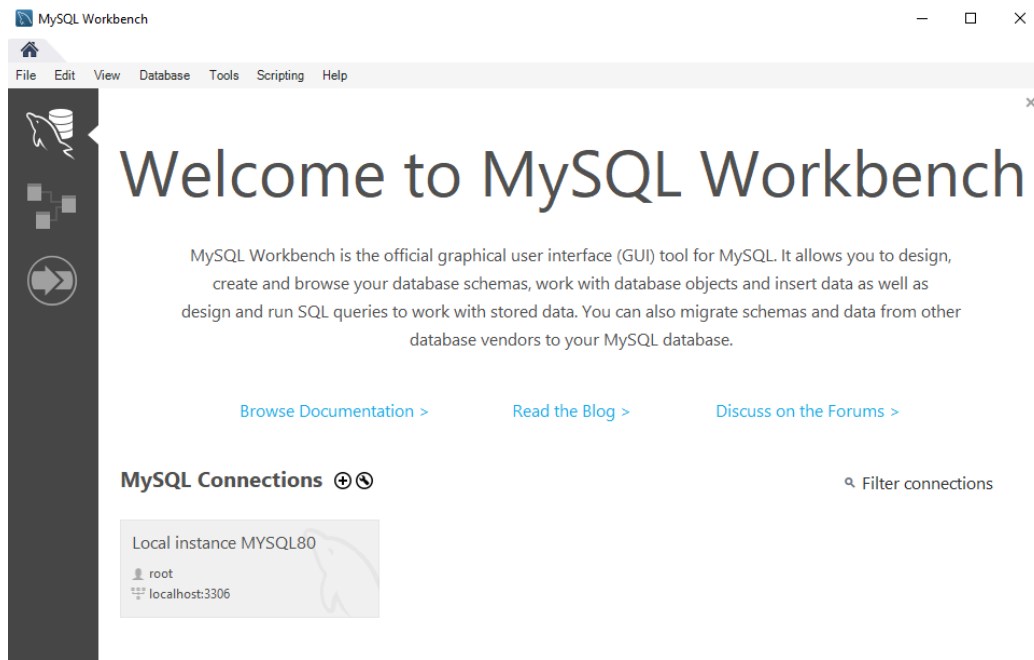
6. Run

```
CREATE USER 'admin'@'localhost' IDENTIFIED BY 'password';  
GRANT ALL PRIVILEGES ON *.* TO 'admin'@'localhost' WITH GRANT OPTION;
```

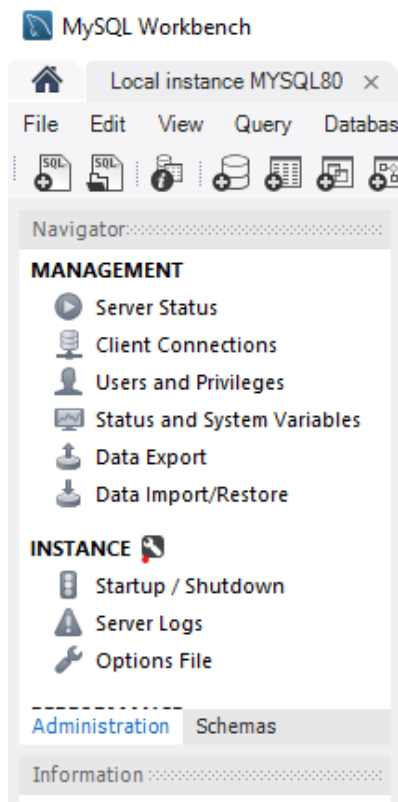
Then you can connect via MySQL Workbench using admin username and password as your password, whichever you have chosen.

Running MySQL with MySQL Workbench

We will be using MySQL Workbench to run SQL queries in this class. Open up the MySQL Workbench program and you should be greeted by this screen:

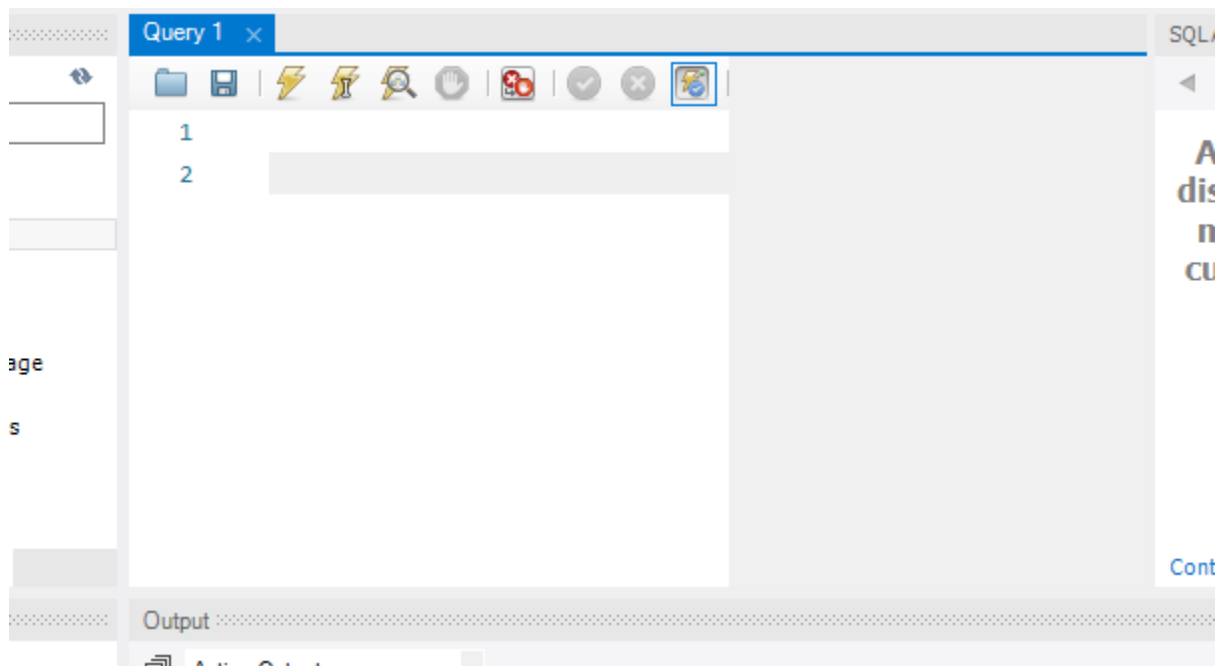


Click on the box under MySQL Connections labeled “Local instance”, where you will be prompted to enter your root password. You will also have the option to save the password (**recommended**). Once



logged in, you will be greeted by the main GUI for interacting with your local MySQL Server. On the left-hand side of the screen, you should see 2 tabs in the middle labeled Administration and Schemas.

Select the schemas tab to see your available SQL databases. By default, you should have schemas pre-installed. Double-clicking on the “barrels” next to the database names will select that database as active. When using a specific database, select the barrel icon next to the database. The name should turn bold when you do so. We have now selected a database and will be able to query it however we see fit. Let’s try it out. The main window of your screen should be a mostly empty textbox with a tab labeled “Query 1” above it:



Select this window, type “select 1 + 1;” (without the quotations), and then click the middle lightning bolt icon to run the query. You should see the results populate in the Output window below. Congrats, you just ran your first MySQL query! This is only the basics of using the MySQL Workbench program, and we will return to it in more detail later in the semester.

Running MySQL on Command Line (Optional)

In this course, we will be focusing on using MySQL through the graphical user interface MySQL Workbench. However, you can also access MySQL through the command line (macOS users please read the **macOS Path Echo Instructions** section below first).

Running the MySQL Interactive Command Line Client

To start, first open up your computer's command line / terminal. To access MySQL through the command line, run one of the following commands:

Command	Description
<code>mysql -u root -p</code>	Access MySQL if you have a server password set up
<code>mysql -u root</code>	Access MySQL if you do not have a server password set up

Loading SQL Scripts

Command	Description
<code>mysql -u root -p < file_name</code>	Use while inside of command line / terminal
<code>source file_name.sql;</code>	Use while inside MySQL server client

MySQL Commands

Command	Description
<code>show databases;</code>	Print a list of databases currently stored
<code>use database_name;</code>	Access and edit a specific database
<code>show tables;</code>	Print a list of tables currently stored in your current database

macOS Path Echo Instructions

For macOS users, you must include the specific path for MySQL in order to run mysql commands in Terminal. For **most** macOS users, you can run one of the following commands:

Command	Description
<code>/usr/local/mysql/bin/mysql -u root -p</code>	Access MySQL if you have a server password set up
<code>/usr/local/mysql/bin/mysql -u root</code>	Access MySQL if you do not have a server password set up

You must include the specific path `/usr/local/mysql/bin/` each time you want to run a mysql command in Terminal.

If you wish to avoid including the path each time you call the mysql command, run the following command in Terminal:

```
sudo sh -c 'echo /usr/local/mysql/bin > /etc/paths.d/mysql'
```

Closeout of Terminal after running this command. When you open Terminal again, you can run mysql commands without including the path, such as `mysql -u root -p`.

Update MySQL Server Password

Note: This section's instructions are used to update your SQL password if and only if you remember your previous password!

Congratulations on installing MySQL! If you know your password and you want to **change your password**, there are 2 different methods of doing so. If you do not know your password, see a TA during office hours.

Using Command Line / Terminal

1. Type in `mysql -u root -p` in your command line and enter the password that you had when initially installing your MySQL server.
2. Once you get in your MySQL Interactive Command Line Client, type in the following command: `alter user 'root'@'localhost' identified by 'newpassword';`

And that should change your password!

Using MySQL Workbench

1. Open up the MySQL Workbench application.

2. Log onto your local instance MySQL server.
3. In the query editor, type in the command: `alter user 'root'@'localhost' identified by 'newpassword';` and run the command.

And that should change your password!